2.9 Dimension and Rank

Coordinate Systems

- The main reason for selecting a basis for a subspace H, instead of merely a spanning set, is that each vector in H can be written in only one way as a linear combination of the basis vectors.
- To see why, suppose $\mathcal{B} = \{b_1, \ldots, b_p\}$ is a basis for H, and suppose a vector x in H can be generated in two ways, say,

$$x = c_1 b_1 + \ldots + c_p b_p \text{ and } x = d_1 b_1 + \ldots + d_p b_p \quad (1)$$

Then, subtracting gives

$$0 = x - x = (c_1 - d_1) b_1 + \ldots + (c_p - d_p) b_p \quad (2)$$

Since $\mathcal{B}$ is linearly independent, the weights in (2) must all be zero. That is,